

```
ubuntu | SUCCESS => {
  "changed": false,
  "failed": false,
  "ping": "pong"
}
```

Pre-requisites

The Graylog software has a few dependency packages that need to be installed as pre-requisites. The APT package repository is updated and upgraded before installing the pre-requisite software packages.

```
---
- name: Pre-requisites
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [prerequisite]

tasks:
  - name: Update the software package repository
    apt:
      update_cache: yes

  - name: Update all the packages
    apt:
      upgrade: dist

  - name: Install pre-requisite packages
    package:
      name: "{{ item }}"
      state: latest
    with_items:
      - apt-transport-https
      - openjdk-8-jre-headless
      - uuid-runtime

- pwgen
```

The above playbook can be invoked as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/graylog.yml --tags prerequisite -K
```

The ‘-K’ option prompts for the sudo password for the ‘ubuntu’ user. You can append multiple ‘-v’ to the end of the playbook invocation to get a more verbose output.

MongoDB

Graylog uses MongoDB to store meta information and configuration changes. The MongoDB software package that ships with Ubuntu 16.04 is supported by the latest Graylog software. The Ansible playbook to install the same is as follows:

```
- name: Install MongoDB
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [mongodb]

tasks:
  - name: Install MongoDB
    package:
      name: mongodb-server
      state: latest

  - name: Start the server
    service:
      name: mongodb
      state: started

  - wait_for:
      port: 27017
```

The Ubuntu software package for MongoDB is called the ‘mongodb-server’. It is installed, and the database server is started. The Ansible playbook waits for the MongoDB server to start and listen on the default port 27017. The above playbook can be invoked using the following command:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/graylog.yml --tags mongodb -K
```

Elasticsearch

Elasticsearch is a search engine that is written in Java and released under the Apache licence. It is based on Lucene (an information retrieval software library) and provides a full-text search feature. The *elastic.co* website provides *.deb* packages that can be used to install the same on Ubuntu. The Ansible playbook for this is provided below:

```
- name: Install Elasticsearch
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [elastic]

tasks:
  - name: Add key
    apt_key:
      url: https://artifacts.elastic.co/GPG-KEY-
elasticsearch
      state: present

  - name: Add elastic deb sources
    lineinfile:
```